

MATH 3260 - Chapter 3 Worksheets

Determinants

Name: _____

Date: _____

These are student practice worksheets. They are designed to be used with the chapter notebook and notes, but every problem below contains the data needed to work it independently. No solutions are included in this student version.

Section 3.1 - The Determinant of a Matrix

Show enough work to make your reasoning clear. Interpret your answer when a problem asks for interpretation.

1. Compute

$$\det \begin{bmatrix} 7 & 2 \\ 3 & 5 \end{bmatrix}.$$

2. For

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 3 & -1 & 4 \\ 2 & 1 & 5 \end{bmatrix},$$

find all minors M_{ij} and cofactors C_{ij} .

3. Use cofactor expansion to evaluate

$$\det \begin{bmatrix} 4 & 0 & 2 \\ 1 & 3 & 0 \\ 0 & 5 & 1 \end{bmatrix}.$$

Before evaluating, write the expansion explicitly as “entry times 2×2 determinant” and show the alternating signs.

4. Find the determinant of the triangular matrix

$$\begin{bmatrix} 2 & 0 & 0 & 0 \\ -1 & 3 & 0 & 0 \\ 4 & 5 & -2 & 0 \\ 7 & 1 & 6 & 5 \end{bmatrix}.$$

Explain why no cofactor expansion is needed.

5. Explain geometrically why a 2×2 matrix with determinant zero collapses area.

Section 3.2 - Determinants and Elementary Operations

Show enough work to make your reasoning clear. Interpret your answer when a problem asks for interpretation.

1. Use row operations to find

$$\det \begin{bmatrix} 3 & 1 & 2 \\ 6 & 5 & 1 \\ 0 & 4 & 7 \end{bmatrix}.$$

Track the determinant effect of every row operation.

2. Suppose $\det(A) = -6$. Find the determinant after each operation:

- (a) interchange rows 1 and 2;
- (b) multiply row 3 by -3 ;
- (c) add 5 times row 1 to row 2.

State the general rule used in each part.

3. Explain why

$$\det \begin{bmatrix} 2 & 5 & 1 \\ 4 & 10 & 2 \\ 3 & 1 & 7 \end{bmatrix} = 0.$$

4. Use a column operation to evaluate

$$\det \begin{bmatrix} 1 & 3 & 2 \\ 2 & 6 & 5 \\ 4 & 12 & 1 \end{bmatrix}.$$

5. A data matrix has a column equal to the sum of two other columns. What can you conclude about its determinant, column dependence, and invertibility?

Section 3.3 - Properties of Determinants

Show enough work to make your reasoning clear. Interpret your answer when a problem asks for interpretation.

1. Suppose $\det(A) = 4$ and $\det(B) = -3$ for 3×3 matrices. Find
 - (a) $\det(AB)$,
 - (b) $\det(BA)$,
 - (c) $\det(2A)$,
 - (d) $\det(B^{-1})$.
2. Find $\det(2AB^{-1})$ using the same information. State where the exponent 3 enters the scalar-multiple rule.
3. If $\det(C) = 0$, state what you can conclude about C^{-1} , the columns of C , and the homogeneous system $C\mathbf{x} = \mathbf{0}$.
4. A two-dimensional transformation first scales signed area by -2 and then by 5 . What is the signed area scale of the composition?
5. If $\det(D) = -2$, find $\det((D^T)^3)$.
6. Let

$$M = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}.$$

Compute $\det(M)$ and $\det(4M)$ directly, then verify $\det(4M) = 4^2 \det(M)$. This matrix is deliberately nontriangular.

Section 3.4 - Applications of Determinants

Show enough work to make your reasoning clear. Interpret your answer when a problem asks for interpretation.

1. For

$$A = \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix},$$

find $\text{adj}(A)$ and use it to compute A^{-1} . Verify the inverse.

2. Use Cramer's Rule to solve

$$\begin{cases} x + y + z = 9, \\ 2x - y + z = 5, \\ x + 2y - z = 4. \end{cases}$$

3. Find an equation of the line through $(1, 4)$ and $(5, 12)$ using a determinant. Rewrite the final result in a familiar line form.
4. Find the area of the triangle with vertices $(0, 1)$, $(6, 2)$, and $(2, 7)$. Explain why a zero determinant would indicate collinearity.
5. Find an equation of the plane through $(1, 1, 0)$, $(3, 0, 1)$, and $(0, 2, 2)$. Identify a normal vector to the plane.
6. Find the volume of the tetrahedron with vertices $(0, 0, 0)$, $(4, 0, 0)$, $(0, 3, 0)$, and $(0, 0, 2)$.